

Singapore **Public Healthcare** **Sustainability**



Introduction

Demographic Projections

Ageing Population

2030: 1 in 4 Singaporeans
will be elderly

Hospital Admissions Forecasting

2035

Increase Hospital Demand

2023: 62.7% of total hospital
admissions were elderly

Healthcare Expenditure

2025: SGD 20.9 billion
2030: > SGD 30 billion

Increase Healthcare Expenditure

Financing Adequacy Analysis

Data Sources



Data	Purpose
Singapore Population Structure	Analyse growth trends and project future population by age
Hospital Admissions (Age, Sex)	Establish hospital admission patterns for forecasting
Bed Occupancy Rates	To determine hospital capacity utilisation and constraints
Government Healthcare Expenditure	Establish spending trend and calculate cost per admission
Household Expenditure Survey	Establish household healthcare expenditure

1. Demographic Projection

Project Singapore's population structure up to 2035 using historical data with life expectancy trends, mortality and fertility rates.

**Population 4.20M (2025) →
4.10M (2035)**

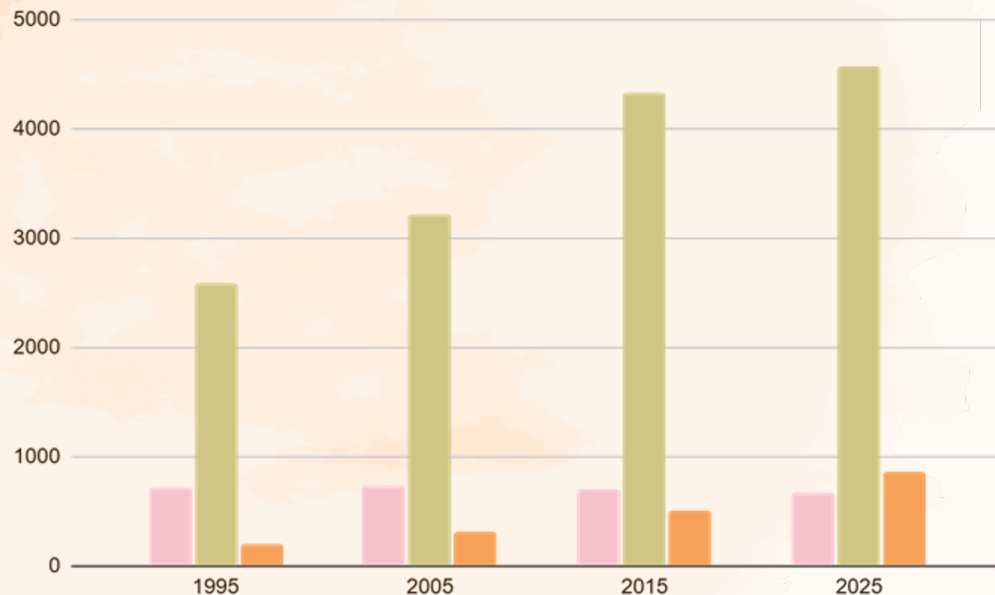
Resident population declining
Resident pop: 4.20M → 4.10M
Net decline of ~107K by 2035

The Elderly



Elderly 25.9% in 2035 (18.8% in 2025)
> 80 years old **doubles to 316K+ by 2035**

- **0-14 years: declining to 12.1%**
- **15-64 years: shrinking to 63.3%**
- **65+ years: rising to 25.9%**



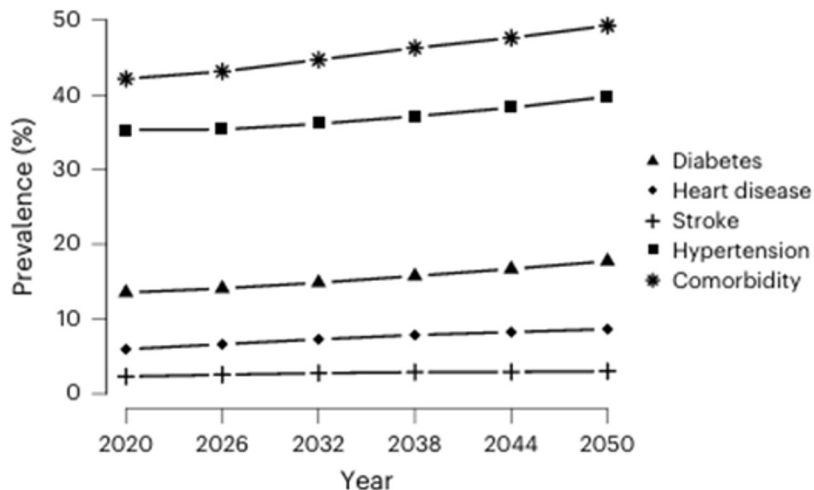
Source: <https://www.population.gov.sg/> & <https://www.singstat.gov.sg/>



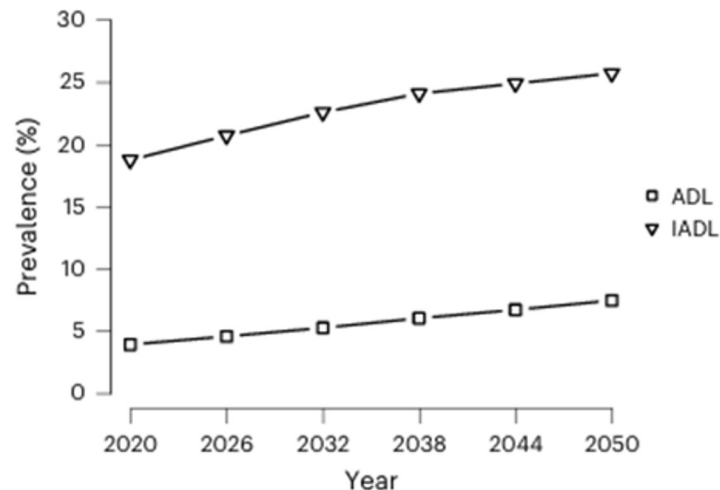
2. Hospital Demand & Admission Forecasting

Project demand for hospital beds using age-specific hospital admission rates to assess if projected demand aligns with planned hospital capacity.

a Chronic disease



b Disability



- **Bed-day demand: 3.79M days**
→ **4.45M days (+17%)**

- **BOR 2025: 89.6% — exceeds 85% red line**

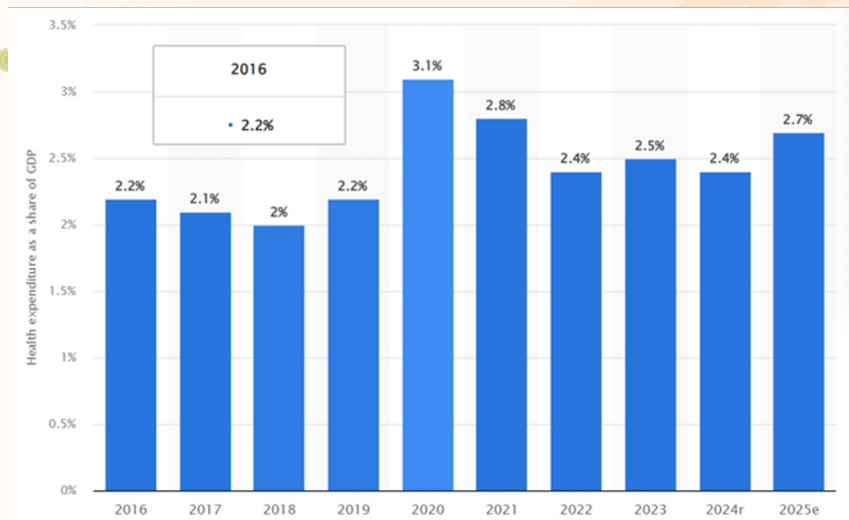
Total admissions grow: 601K (2025) → 685K (2035)

Elderly (65+) admissions: 266K → 349K (+31%)



3. Healthcare Costs Projection

Healthcare expenditure is driven by changes in demand and medical cost inflation.



Govt Hospital Opex

2025: SGD 7.52B → 2035: SGD 14.10B
(+88%)

Medical Inflation

Rising 2.7% (2025) to 6.1% by 2035

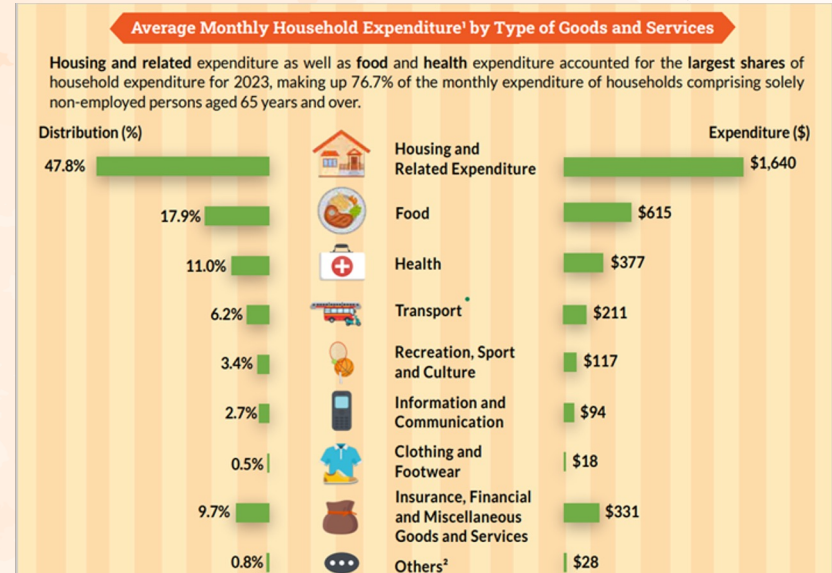
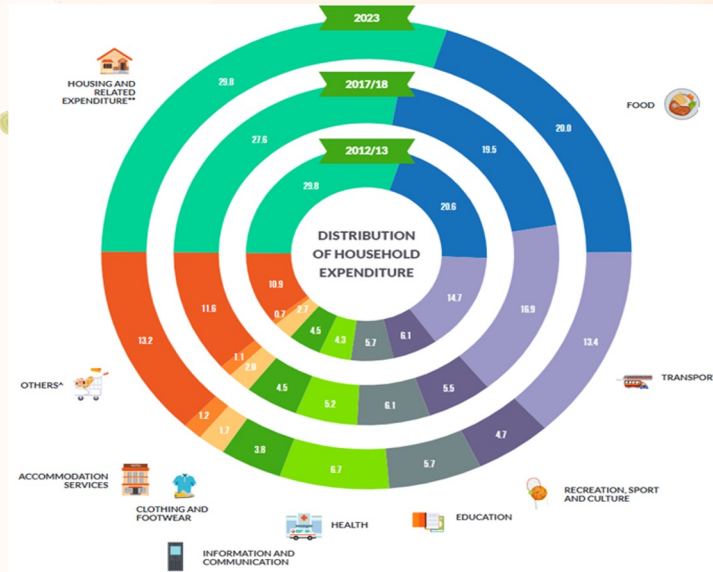
Cost Per Admission

SGD 12,490 (2025) → SGD 17,644
(2035)

● **Elderly cost multiplier drives accelerating expenditure**

● **Projected Govt Healthcare Cost 2026–2035**

4. Financing & Subsidy Sustainability



Financing gap emerges; surplus until 2030, deficit from 2031

Deficit reaches SGD 951M by 2035 (7.2% of govt budget)

Household OOP burden: SGD 1,760/yr (2026) → SGD 2,698/yr (2035)

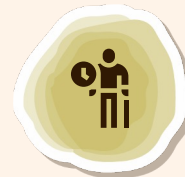
Source: <https://www.singstat.gov.sg/find-data/search-by-theme/households/household-expenditure/latest-data>

Trade Off Analysis



Subsidy Level vs Fiscal Sustainability

- Trade-off between household protection and fiscal sustainability
- 65% subsidy: financing deficit SGD 951M by 2035; OOP SGD 4,358/yr per household
- 55% subsidy: deficit eliminated; but OOP jumps to SGD 5,603/yr (+29%)
- Optimal zone: 60% subsidy — deficit SGD 134M, OOP SGD 4,980/yr (21.6% of income)



Capacity Expansion vs Fiscal Sustainability

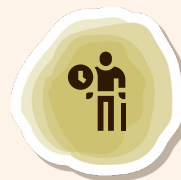
- Balances operational capacity needs with fiscal constraints
- Baseline 3.53% expansion rate: BOR falls from 89.6% (2025) to 74.2% (2035)
- 2% expansion rate: BOR stays above 86.8% through 2035, breaching safety threshold
- 3.53% expansion rate (MOH plan) keeps BOR within 85% threshold by 2032

Sensitivity Analysis



Medical Cost Inflation vs Government Health Subsidies

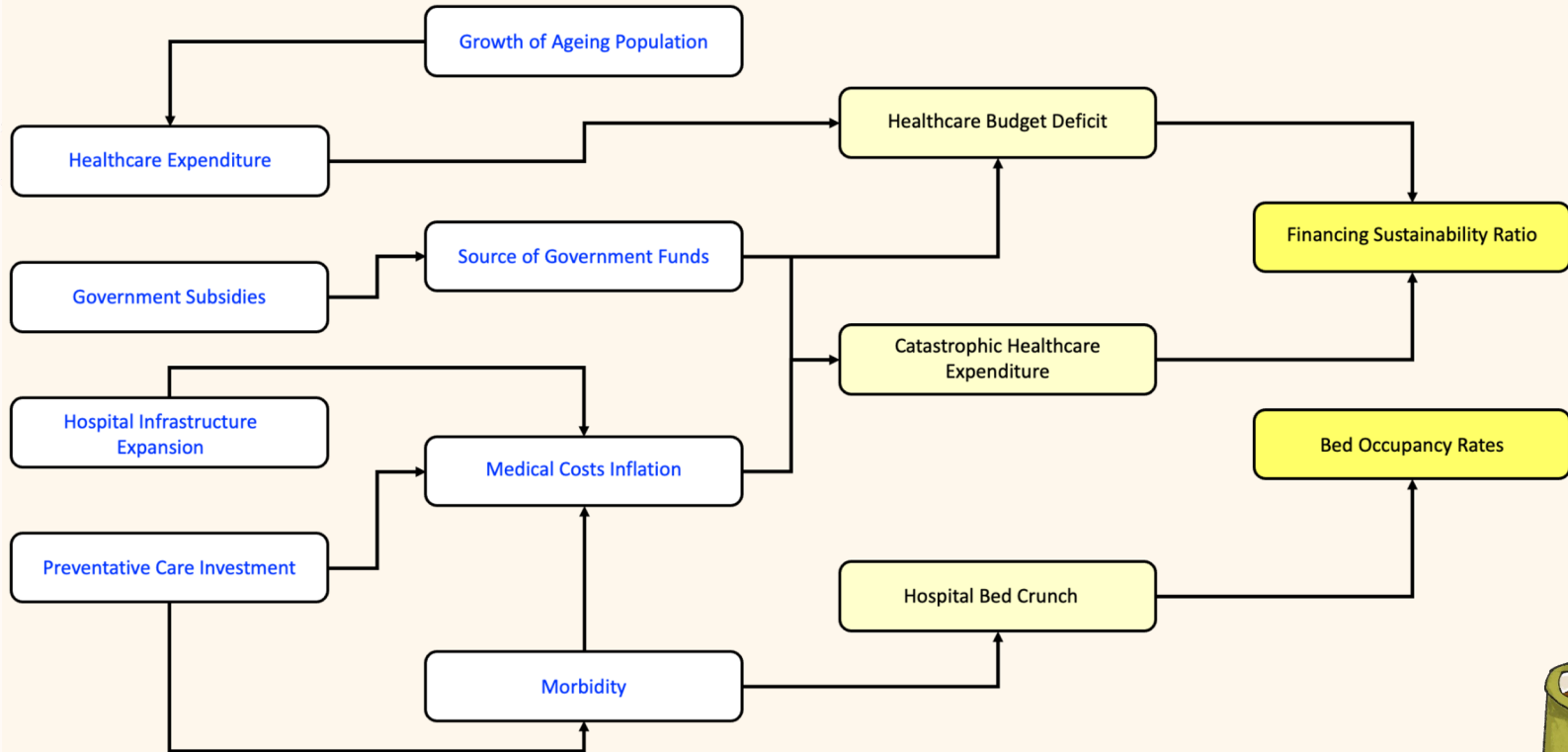
- At 2.7% inflation (base): deficit SGD 951M by 2035
- At 5% inflation: deficit balloons to SGD 3.6B by 2035
- At 8% inflation: deficit reaches SGD 9.6B — fiscal crisis threshold
- Each 1% rise in inflation adds ~SGD 700M to 2035 deficit



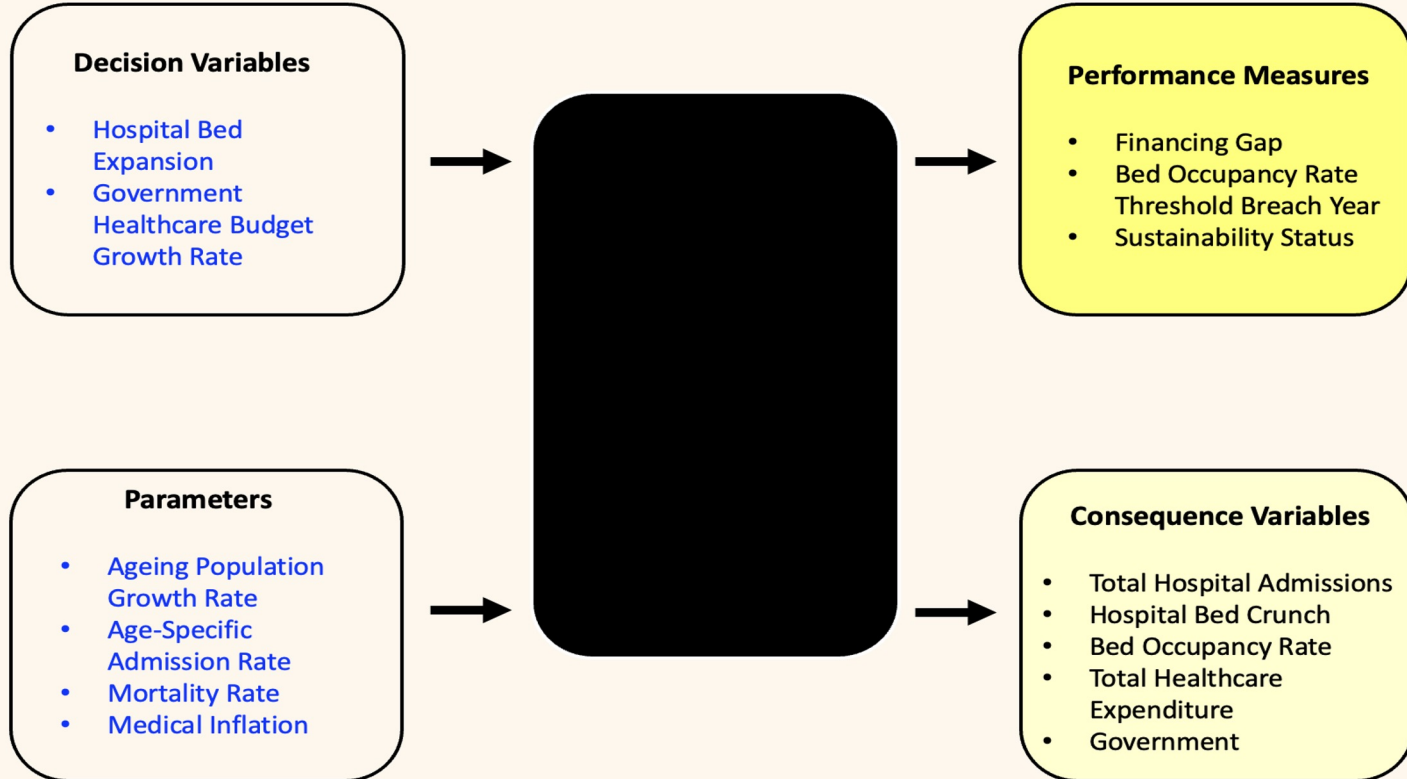
Elderly Population Growth Rate vs Projected Hospital Admissions

- Bed expansion rate is the #1 BOR driver: 3.53%→2% raises BOR by 12.7pp
- ALOS reduction: 10% ALOS cut reduces BOR by ~5.5pp — policy lever
- 65+ population growth: 1.5% faster growth adds only 0.68pp to BOR
- Healthier SG policy factor: -0.6% efficiency gain reduces BOR by 3.3pp

Influence Diagram



Black Box Model



Key Findings & Recommendations

Hospital Capacity

Demographics → admissions
→ BOR → expenditure
(2026–2035)

Subsidy Optimisation

60% subsidy rate balances
household burden and govt
fiscal position



Fiscal Sustainability

Deficit of SGD 951M by
2035; sustain budget
growth at 5.75% p.a.

Policy Priority

Prioritise Healthier SG, ALOS
reduction, and 3.53% bed
expansion plan



Thank you!